

Aayush Kumar Jha

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Education

Indian Institute of Technology, Dharwad

B.Tech, Mathematics and Computing, CGPA-9.48

Dharwad, India

2024 – 2028

Core Coursework: Data Structures & Algorithms, Computer Architecture, Graph Theory, Linear Algebra, Probability Theory & Data Analysis, Discrete Structures, Artificial Intelligence, Machine Learning, Deep Learning, Natural Language Processing

Projects

Satellite Imagery Super-Resolution (SIH 2026)

August 2026 – Present

Python, PyTorch, FastAPI, OpenCV, SQLite

- Engineered an end-to-end 4x Super-Resolution pipeline for 4-channel (RGB+NIR) Sentinel-2 satellite imagery using a custom 12-block RRDBNet and PatchGAN discriminator.
- Eradicated adversarial mode collapse and checkerboard artifacts via ESRGAN-style Network Weight Interpolation, achieving 36.94 dB PSNR and 0.867 SSIM on held-out test data.
- Deployed the hardware-optimized model via a localized FastAPI backend integrated with a Glassmorphism web dashboard, featuring an interactive split-viewer and persistent history.

Underwater Image Enhancement using CNN Autoencoder

March 2026-April 2026

Python, TensorFlow, OpenCV

- Built a convolutional autoencoder with skip connections to restore color and clarity in underwater images, trained on 712 raw-reference image pairs.
- Achieved a Mean PSNR of 18.96 dB and Mean SSIM of 0.82 across 178 held-out test images using a ~3.5M-parameter model.
- Trained with AdamW optimizer and Huber loss, using a ReduceLROnPlateau scheduler over 180 epochs on an end-to-end preprocessing-to-evaluation pipeline.

ToyRISC Processor Simulator

January 2026-April 2026

Java, Computer Architecture

- Designed and implemented a cycle-accurate simulator for a custom ToyRISC instruction set as part of a multi-lab Computer Architecture project.
- Built modules for instruction fetch-decode-execute, register file, and memory access, progressively extending the ISA across six iterations.
- Validated simulator correctness against expected register/memory states for a suite of custom assembly test programs.

A2A TALENTVector (HR Talent Search Engine)

March 2026

Architect & Model Designer | Nasiko AI Agent Buildathon

- Engineered an autonomous talent sourcing system using Google A2A Protocol and JSON-RPC 2.0.
- Integrated third-party OAuth for secure email and profile data access.
- Developed a Log-Normalized Multi-Metric Scoring Engine to rank developers and mitigate outlier bias.
- Built an automated data parsing pipeline utilizing GPT-4o APIs for robust system integration.

Computer Vision & YOLO

May 2026 – Present

- Engineered and deployed custom YOLO object detection models via Python scripts.
- Managed end-to-end data pipelines with CVAT.ai labeling and Roboflow dataset integration.
- Deployed and evaluated models locally to ensure high-performance execution.

Low-Resource Neural Machine Translation: en ↔ mni

May 2026 – Present

- Fine-tuned 1.1B/4.3B models (IndicTrans2, Sarvam) for Manipuri using LoRA/DoRA, achieving 51.55 BLEU.
- Built end-to-end MT pipeline for data quality analysis, data splitting, and multi-metric evaluation.
- Implemented data augmentation via back-translation, expanding training data by ~2.8x.
- Analyzed Seq2Seq vs CausalLM architectures with LoRA/DoRA adapters on an RTX 6000 Ada GPU.

Publications & Academic Research

Co-Author | “Cryptographic Collapse Forecasting: A Strategic Shift to Quantum Security”

September –

December 2025

32nd IEEE International Conference on High Performance Computing, Data, and Analytics (HiPC 2025)

- Collaborated with FutureG Labs, IIT Dharwad on Quantum-secured networking within O-RAN architectures.
- Presented findings via a Poster Presentation to international domain experts in Hyderabad.

Achievements

- **Academic Excellence:** Maintained a 9.48 CGPA in the Mathematics and Computing department.
- **Buildathon Participant:** Successfully developed an autonomous agent at the Nasiko AI Agent Buildathon.

Coding Profiles & Codebases

- [LeetCode Profile](#)
- **Java Data Structures & Algorithms Codebase:** A complete, self-authored repository featuring robust implementations of fundamental and advanced data structures and algorithms written entirely in Java.

Technical Skills

- Languages:** Core JAVA, C, Python, Bash.
- Computer Science Fundamentals:** Data Structures and Algorithms, Object-Oriented Programming, Graph Theory, Computer Architecture, Operating System.
- Artificial Intelligence:** Machine Learning, Deep Learning, Natural Language Processing
- Tools & Technologies:** VS-Code, Git, Docker, LLM API Integration, Logarithmic Modeling, Statistical Normalization, LaTeX.

Leadership & Extracurricular

Member | Software and Mechanical Team, Robotics Club *September 2024 – May 2026*
IIT Dharwad

- Collaborating with a multidisciplinary team to design and develop a SLAM (Simultaneous Localization and Mapping) Robot.
- Contributing to software implementation and mechanical design to enable autonomous navigation capabilities.

Student Mentor *Jul 2025 – May 2026*
Student Mentorship Program, IIT Dharwad

- Provided academic guidance to first-year students, and conducted regular group sessions for them.
- Offered support and guidance to help students transition to college life.